Do longer housing searches lead to better matches? Evidence from the voucher program*

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Abstract

Low-income households often struggle to find affordable housing that meets their needs. While the Housing Choice Voucher program helps over 2.3 million households per year afford rental housing, participants in the program still face heightened challenges in using their vouchers, finding stable housing, and reaching higher opportunity neighborhoods. In this paper, I investigate the role of search time in promoting better housing matches within the voucher program. Using features of the random timing of voucher issuance, I show that there is no meaningful effect of issuance timing on the probability of using a voucher (in other words, the probability of making any match), but there is a strong effect on the length of time that households search for when they do use their vouchers. I then use issuance timing as an instrument for search time, and find that longer searches are associated with declines in the probability of exiting the program or moving housing units within 2 years of entering. Longer searches also allow households to reach lower poverty neighborhoods than baseline. Effects are strongest in tighter rental markets, where search costs tend to be highest.

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1 Introduction

In increasingly tight housing markets, low-income households often struggle to find safe and affordable rental housing that meets their needs. Due to a combination of rising rents, stagnating incomes, and a lack of new affordable homes, more than half of all US renters were cost burdened in 2022, paying more than 30 percent of their income in rent, an all time high (Joint Center, 2024). There are certain groups in particular that face heightened challenges in the housing search process, including Black and Hispanic households (Christensen et al., 2021; Christensen and Timmins, 2021; Arefeva et al., 2024), and households with children (Aron et al., 2016).

Another group that may struggle in the housing search process is voucher recipients. The Housing Choice Voucher program subsidizes over 2.3 million households annually; however, voucher recipients have to find and rent homes on the private rental market to receive their subsidies. In additional to contending with a shortage of affordable homes, voucher recipients have several additional challenges. First, they have a limited time to search for housing in order to receive their benefits. They also must find housing units that adhere to all program guidelines. Voucher recipients also must find a landlord who is willing to rent to them and participate in the program, which can be a high hurdle to clear as landlords often discriminate against voucher holders (Aliprantis et al., 2022; Faber and Mercier, 2022; Cunningham et al., 2018; Freeman and Li, 2014; Phillips, 2017).

The consequences of the difficult housing search process are reflected in programmatic outcomes. Most fundamentally, in recent years only 60% of households issued a voucher are able to use that voucher at all. Even for households who are successful in using their vouchers, there is still evidence of high search costs, which can be measured by looking at search time, or the elapsed time between the issuance of a voucher and a participant successfully using their voucher to lease a home. Search times increased sharply between 2018 and 2021, now standing at around 75 days, which is longer than the federal minimum 60 days that housing authorities must give households to use their vouchers. Additionally, despite the voucher program's core programmatic goal of opening up more neighborhoods to program participants, households with vouchers often live in similar neighborhoods to unassisted, low-income renters (Jacob et al., 2015; Galvez, 2011; Pendall, 2000). Low success rates, growing search times, and constrained locational choices can all be reflections of the difficulty of the housing search process for voucher recipients.

In this paper, I study the effect of search times on housing match quality in the voucher program. Though there is wide variation in search times both across and within the 2,000+ housing authorities who administer the program across the country, there is very little research that looks at the drivers of this variation or the effects of longer searches. Specifically, I evaluate how longer searches can lead to lower turnover, in terms of program exits and moving units, and the ability of households to reach lower poverty neighborhoods.

I use novel administrative data to measure success rates and search times for over 600 housing authorities across the country, which allows a more detailed analysis of these programmatic outcomes than past studies. The empirical strategy is derived from the fact that the timing of voucher issuance is random, but there are features of issuance timing that can affect the housing search. Vouchers can be issued on any day of the month, but households generally can only lease-up at the beginning of the month. This mechanical relationship means that those issued a voucher at the end of the month have more time to search, as they generally roll over and start their leases one month later. The month of issuance also impacts the search process, both through seasonal variation in rental markets, and housing authority practices around search times. This source of random variation in search time allows for the causal identification of the effect of longer searches.

First, I test the effect of the day or month of voucher issuance on the probability of successful voucher use- in other words, the effect on making any match. Given the result that the timing of voucher issuance does not affect the probability of successful lease-up, I use the timing of issuance as an instrument for observed search time to test, for participants who are successful in using their vouchers, the effect of longer searches on several different margins of match quality. I also conduct heterogeneity tests, motivated by the fact that searches tend to be longer for certain groups, including households of color, and are considerably longer on average in tighter markets.

I find that longer searches lead to increased program retention, and a longer lengths of stay in the same housing unit. Specifically, a 50 percent increase in search time leads to 4.8 percent decline in leaving the program within 2 years, and a 2.8 percentage point decline in moving units within two years. These results are primarily driven by Black households and households with children.

Longer searches also allow voucher recipients to reach lower poverty neighborhoods. Increasing search time by 50 percent leads to a 2.5 percentage point (7%) increase in the chance of moving to a ZIP Code with a lower poverty rate than baseline. These results are also predominately driven by Black households and households with kids, and, unlike the effects on housing turnover which are prevalent across housing market characteristics, are entirely driven by tighter rental markets.

This paper contributes to several strains of literature. First, these results draw important parallels to other literature on search and match quality. Perhaps most closely related is the job search literature, where several studies show that longer job searches can lead to better matches, predominately measures by longer stays (Belzil, 2001; Caliendo et al., 2013; Ehrenberg and Oaxaca, 1976).

Second, this paper contributes to the literature on housing search. While there is a large body of work on housing search, much of the quantitative evidence focus on the home-buying process. Several studies show differences in the housing search process by income, race, and type of constraints (Harvey et al., 2020; Krysan and Crowder, 2017). This paper also adds to a body of predominately qualitative work on the search process for voucher recipients (Garboden, 2021; McCabe, 2023; Rosen, 2014, 2020), and a small but growing set of work on success rates and drivers of success rates within the voucher program (Ellen et al., 2024; Chyn et al., 2019; Finkel and Buron, 2001). The results on neighborhood quality also tie into research on housing mobility programs, and reveal a potentially low or no-cost method of expanding neighborhood choice in the voucher program (Bergman et al., 2024; DeLuca et al., 2023).

Third, this paper adds to a growing literature on administrative burden, and how features of a program's design can affect take-up and retention. Homonoff and Somerville (2021) show that the timing of certain milestones within a program can influence program retention. Several other papers show that features of program design can affect take-up and other programmatic goals through the administrative burden channel (Gray, 2019; Ganong and Liebman, 2018). This paper adds to this literature by providing new evidence on a potential strategy (extending search times) to boost take-up and retention in a program that involves a complex interaction with a private third party.

The paper is organized follows. Section 2 details the institutional context, including background on success rates and the housing search process within the voucher program. Section 3 describes the theoretical argument on how longer searches can affect different margins of match quality. Section 4 describes the data used for the study. Section 5 describes the empirical design. Section

6 provides descriptive information on search and matching within the voucher program. Section 7 shows causal results on the effect of issuance timing on success, and search time on housing turnover and neighborhood outcomes, as well as shows important heterogeneity in results across households and markets. Section 8 discusses robustness, and Section 9 concludes.

2 Institutional context: Housing Choice Voucher Program

2.1 Background on the voucher program

The Housing Choice Voucher program is the largest federal rental assistance program in the country. In 2022, more than 5 million individuals in 2.3 million households received vouchers. While the program is funded federally by the Department of Housing and Urban Development (HUD), it is administered locally by more then 2,000 public housing authorities (PHAs). PHAs have discretion over the daily operation of the program, but abide by certain federal guidelines set by HUD.

Vouchers can be transformative for low-income households who are able to use them. Once a household enters the voucher program, they generally pay 30 percent of their income towards rent, and the local housing authority covers the rest, up to a certain subsidy limit, called the payment standard. Rent burdens in the voucher program are significantly lower than those of unassisted, low-income renters. Vouchers have also been shown to reduce homelessness, reduce over-crowding, and improve children's performance in school (Schwartz et al., 2020).

However, the program struggles with two main issues. This first is that many households who receive vouchers are not able to use them. In recent years, over 40 percent of households who receive vouchers must return them to the housing authority as they are unable to find and rent a qualified home within the allotted time they are given. Second, the voucher program has not consistently achieved its core goal of allowing voucher holders to live in a diverse set of neighborhoods. Both of these issues reflect the difficulty that voucher recipients may have during the search process.

2.2 Take up in the voucher program

Take-up in the voucher program is a multi-step process, often spanning multiple years. First, voucher holders must apply to join a waitlist at a PHA. Many PHAs will close their waitlists due to outsized demand for the program, so in order to even apply, participants must often wait for the PHA in their local market to open their wait list (Hembre and Urban, 2023). At this point, households apply, and if they are certified to qualify for the program, they are placed on the wait list. While wait times vary widely across the country, they are generally quite long. On average, households wait 2.5 years, but at some housing authorities included in the sample for this paper, wait times are as high as 8 years (Acosta and Gartland, 2021).

Once a household is selected off the waitlist, they are contacted by the PHA, confirm their eligibility and interest in the program, and provide documentation. Only then are they issued a search voucher. A search voucher is a promise of receiving the subsidy, given the voucher recipient is able to secure a qualifying unit. At this point, households must be given a minimum of 60 days to search, though many PHAs offer longer search windows. PHAs also have discretion in granting extensions. Many PHAs offer some sort of automatic extensions, and virtually all PHAs grant extensions for a wide variety of acceptable reasons.

Households now must find a unit that meets all programmatic requirements. The unit must have a rent that falls within the payment standard, and must pass a housing quality inspection.[1] The landlord for the unit must also be willing to rent to a voucher holder and participate in the program.

Vouchers are not an entitlement, which makes calculating a traditional take-up rate difficult. Only one in five low-income households with unmet housing needs get any form of housing assistance (Kingsley, 2017). In this paper, I focus only on eligible households who are issued a voucher, a small subset of all eligible households. The share of households who are able to use their vouchers once they are issues a voucher is called the success rate; a search is considered successful if the household is able to use the voucher within one year of issuance.[2] It's important to note that at this point in the process, households have repeatedly expressed interest in the program and confirmed their eligibility. The voucher program also provides quite generous subsidies. The average household in the sample for this paper received $8,000 per year, and in more expensive markets, this number

[1]Payment standards are based on HUD's Fair Market Rents (FMRs). PHAs have discretion to set payment standards within 90-110% of FMRs. Voucher holders may rent units above the payment standard, but they are required to pay the difference between the rent of the unit and the payment standard out-of-pocket, and when a voucher recipient joins the program, their rent payment cannot exceed 40%.

[2]The one year window allows for comparable measures across years. Most searches that are successful will conclude within a year; in the sample for this paper, 99 percent of successful searches ended within a year.

can be substantially higher. This makes low, and declining, success rates even more striking, and invites further examination of the housing search process.

The discretion that PHAs have over search time policies is important, as it's an lever that PHAs can use to affect success rates and other outcomes for tenants. In the Housing Choice Voucher Guidebook, HUD states that search times should be monitored and evaluated for best outcomes; however, this has never been done on a large scale due to a lack of representative and detailed data. Although differences across PHAs are held constant at this point, this paper provides some of the first insight on how search time can affect outcomes for voucher holders, an important consideration for housing authorities and policymakers.

3 Theory

3.1 Factors affecting the search process

In general, low-income household often face a difficult housing search process. Evidence shows that low-income households who are more constrained in their search often have different goals from higher income households; while higher income households search for "forever homes" that will meet their long-term needs, lower income household often end up accepting homes that they view as "temporary stops" and sacrificing on some of their long term goals and preferences (Harvey et al., 2020). These differences can often be exacerbated further when households are forced to move, which is particularly relevant for new voucher recipients who have only a limited time to use their voucher.

The voucher program predominately assists Black and Hispanic households. Around 50% of voucher households contain children, with over 90% of those households headed by a single mother. The demographics of the voucher program mean that participants may face confounding challenges. Households of color have been consistently shown to face discrimination in both rental and homeownership markets (Christensen et al., 2021).

In addition to the challenges discussed above, voucher recipients have additional requirements that they have to navigate. While they do not face a traditional budget constraint (voucher holders are indifferent on rent, up to the applicable payment standard), they have to meet several other conditions that can cause many potential voucher holders to either fail to use their voucher altogether, or fail to meet some of their preferences. The first is finding units that are eligible for the program. Cunningham et al. (2018) find that across 5 cities, only 1 in 39 rental units were eligible for the program, given subsidy limits.

The second is finding a landlord that will accept vouchers. Several studies have documented both direct and indirect discrimination against voucher holders. Source of Income (SOI) discrimination laws prohibit landlords from discriminating against potential tenants on the basis of using a voucher. While many states and localities have enacted such laws, there is still frequent discrimination against voucher holders (Cunningham et al., 2018). Landlords are often hesitant to house voucher recipients both because of preconceived notions about voucher recipients themselves, and due the added administrative burden of complying with program requirements, passing housing quality inspections, and working with housing authorities (Garboden et al., 2018). Blanco and Song (2024) find that landlords in higher rent areas are more likely to use negative language towards voucher holders when advertising their units, indicating that it might be even more difficult to find landlords who will accept vouchers in a broader set of neighborhoods. Research showing that voucher recipients reach lower poverty neighborhoods where SOI laws are present also provide evidence of the barrier that landlords can impose on the locational choices of voucher recipients (Freeman and Li, 2014; Ellen et al., 2022).

Voucher recipients may also struggle with the costs associated with administrative burden. Learning, compliance, and psychological costs may all be at play during the housing search process (Herd and Moynihan, 2018). Voucher recipients have only a short time to learn about all program requirements, and understand what these requirements mean in their local context. And although at this point in the process, voucher recipients have already confirmed their eligibility, there are still several administrative steps they must take to find an eligible unit, get it approved, and sign a lease. Psychological costs may also affect the housing search, both due to fear of rejection from landlords in a wider set of neighborhoods (DeLuca et al., 2023), stigma associated with using a voucher, and the difficulty of moving further from existing homes and social networks.

3.2 The role of search time

There are several reasons to believe that voucher recipients who search for longer may be able to find a better match. In the literature on job search, a common finding is that longer searches can lead to a better match, most commonly measured by the length of stay at the next place of employment. The theory underlying these models is that given a shorter time constraint, workers may accept a job that they would otherwise reject, had they been given more time to search. This is frequently evaluated in the context of unemployment insurance, where an important feature of policy design is the benefit duration. This has strong parallels to the search process within the voucher program, where recipients are given a limited duration of time to search for a unit.

Longer searches may also allow participants more time to expand their search beyond their immediate neighborhood. Moving costs can be substantial; perhaps particularly for the large share of voucher households that contain school-age children. Additional flexibility in timing can make it possible for households to search outside of their neighborhood, and ensure that they are able to make the move to a new area within the allotted time.

Additional time can also provide voucher recipients with a better understanding of program requirements, helping to combat the high compliance and learning costs that the voucher program entails. Several papers shows that compliance and learning costs can depress program take-up, and allowing for additional time and support in navigating these costs can have large benefits (Homonoff and Somerville, 2021; Chetty et al., 2013; Bettinger et al., 2012; Barr and Turner, 2018; Armour, 2018).

4 Data

4.1 Administrative data on voucher use

I use detailed administrative data from the Office of Public and Indian Housing (PIH), the division of HUD that is responsible for administering the voucher program. PHAs submit data on each programmatic action to the PIH Information Center (PIC). These actions indicate when each household is issued a voucher, and if and when they are successful in using their voucher. Importantly, I observe key demographics, including race, ethnicity, age, household composition, and origin ZIP Code at the time of voucher issuance. For households who are successful, once they are admitted to the program, I observe their full demographics, complete address, and detailed information on their housing costs and subsidy payments.

For households who are successful in using their vouchers, I also observe all subsequent actions, including moves, ports to other PHAs, income re-certifications, and program exits. From these additional actions, I can construct a history of program participation, and identify how long they remain in the voucher program and at a specific address.[3]

Neighborhood outcomes are derived from merging this administrative data with data from the 2015-2019 American Community Survey (ACS) on poverty rates and poverty percentiles within a CBSA at the ZCTA level. I identify a dichotomous indicator for moves where the destination poverty percentile is higher than the origin poverty percentile; I also use a continuous indicator for the change between the origin and destination poverty percentile.

I use several supplementary data sources to conduct heterogeneity analysis. I use county rent levels from the 1-year 2019 ACS to stratify the sample into low and high rent markets. I also use data on vacancy rates from HUD and the United States Postal Service as an additional method for stratifying the sample based on rental market conditions.

4.2 Sample

The administrative data greatly expands the ability to measure success rates and search times. However, these data have not historically been used to measure these outcomes, and required extensive cleaning. In order to get accurate estimates of these two metrics, I employ data quality standards that select only PHAs where data on voucher issuance appears to be complete (Ellen et al., 2023).

The sample for this paper includes 663 PHAs from the 300 largest CBSAs, in terms of number of voucher holders.[4] Combined, these PHAs cover nearly 70 percent of voucher holders. In general, these PHAs are very reflective of the program as a whole.

Table 1 reveals important characteristics of the sample. Just over half of households are Black; around 15 percent of households are Hispanic. Half of household include children, with the large majority (90%) of households with kids headed by women.

[3]Specifically, I detect changes in addresses by geocoding each unique address entry. The geocoding match rate is above 95%. I then use the cleaned addresses to detect any changes in address. This methodology ignores moves within the same building, so a move is defined as a change of street address, not simply a change in unit. Changing units within a building is relatively rare; I estimate that only around 3% of moves are within the same building.

[4]I exclude 39 legacy Moving To Work (MTW) PHAs, as these PHAs do not report the ZIP Code of origin.

5 Empirical strategy

The effects of search time are difficult to identify, given potential endogeneity issues. Part of this endogeneity comes from unobserved characteristics of households who search longer, which can also affect their immediate and longer term housing outcomes. In this paper, I will distinguish between households who take a long time to complete their housing search, and households who will search for longer when offered additional time.

Another potential confounding factor is that much of the variation in search time across housing authorities may be explained by differing search term and extension policies. Without complete information on search time policies, it is difficult to account for differences in the amount of time that households are given to search. For that reason, I use PHA by year fixed effects, to investigate the effect of observed search time, holding allowable search time constant. The effects can therefore be interpreted as the effects of variation in search time within a PHA.

The basis for the empirical design is that the timing of search voucher issuance is random, but that there are elements of issuance timing that can affect search time and the search process in general. First, the day of the month that a voucher is issued can affect the length of time that a household has to search, since vouchers can be issued any day of the month, but leases generally only start on the 1st. Given a certain search window, this means households at the end of a month have a shorter period of time to start a lease before the search window elapses. In practice, this means that searches tend to be longer when vouchers are issued at the end of the month, since PHAs will give households the flexibility to roll over into an additional month; however, this could also affect success rates if PHAs choose not to exercise this leniency.

The timing of voucher issuance across months can also affect the search. Housing markets are seasonal, but there is also unique variation in searches within the voucher program, beyond seasonal variation. Searches are longest in the summer months, but peak again in November and December; this may be due to reduced PHA capacity to conduct housing quality inspections and approve units. This means that households again can receive additional search time, through the channel of PHA leniency, at certain times of the year.

This design relies on the assumption that timing of voucher issuance is random. Several facts support this assumption. First, most households wait for many years to receive their voucher; the average household waits around 2.5 years, and some households wait significantly longer, upwards of 10 years. Additionally, waitlists are frequently closed, so in practice households all enter the waitlist on the same day. This means that participants cannot manipulate their date of issuance.

While PHAs have different prioritization strategies, these do not necessarily align with the calendar year. PHAs may work through their waitlists with certain strategies, but the timing of waitlist opening and closing varies widely across the country, meaning the households are issued vouchers throughout the month and across months. Figure A1 shows that vouchers are issued throughout the days of the month and across the months of the year.

I first estimate the effect of voucher issuance timing on the probability of using a voucher, and the length of time that it takes for successful households to lease up. The equation is as follows:

$$Y_{itm} = \beta \sum_{j=1}^{12} \mathbb{I}(m - j = 0) + \ \gamma X'_{itm} + PHA_{itm} \times Year_t + \varepsilon \tag{1}$$

where j is an indicator for each month of the year, m is the month of issuance for household i in year t, X' is a vector of individual covariates including race, ethnicity, age, household composition, disability status, and homeless status. Models importantly include PHA by year fixed effects, which means that PHA policies around allowable search times are held constant. These models are estimated using successful voucher use as the dependent variable, and with the log of search time as the dependent variable for only searches that are successful.

A second question is how allowable search time affects other outcomes for households who are successful in their search. If there is no effect of issuance timing on success rates, there is no selection into the sample of successful searches based on timing, and I can then use issuance timing as an instrument for search time. These models take the following two-stage least squares form:

$$Y_{itm} = \beta \log(\widehat{Searchtime}) + \ \gamma X'_{itm} + PHA_{itm} \times Year_t + \varepsilon \tag{2}$$

where the first stage is given by equation 1, using the log of search time as the dependent variable.

In additional to the assumptions detailed above, the standard IV assumptions apply. Figures 5 and 4 show that search time varies across days within a month and months within a year, which serves as the first stage for the IV models.

The exclusion restriction is also satisfied. The month or day of search voucher issuance is unlikely to be correlated with the outcomes of interest (neighborhood poverty rates and housing turnover), except through the effect on search time.

One benefit of using an instrumental variables approach is that the local average treatment effects are a useful tool for thinking about how search time can be used as a tool to improve programmatic outcomes. The LATEs in this study represent the effect of additional search time for households who will "take up" extra search time when their issuance timing allows; therefore, results are informative on how search time policy changes can affect match quality.

6 Descriptive facts about success rates and search times

Low success rates have long been a key challenge for the voucher program, but until recently there was very little insight into voucher success rates or search times. The administrative data used for this paper is one of the only methods that allows for measuring both how many households are able to use their vouchers, and how long this process takes. In this section, I present several descriptive facts about success rates and search times that motivate the following causal analysis.

Prior to 2021, success rates (defined as the share of households who are able to use their voucher within 1 year of issuance) hovered around 60%, but have since declined sharply. By 2022, just over 55% of those issued a voucher were able to lease up and enter the program. At the same time, search times have been sharply increasing. In 2022, it took the median voucher recipient around 75 days to use their voucher, up more than two weeks from two years prior. Both of these metrics indicate growing search costs for voucher recipients.

There is also wide variation in search times across the country (Figure 2. Differences in search time are due to differences in PHA policies, differences in market conditions, and how PHAs respond to differences in market conditions. We see evidence of this in Figure A5 as well, which shows that searches are considerably longer in tighter rental markets, measured by median county rents.

There are also large differences in search times by race and ethnicity, and these difference persist even when we take into account differences in household composition. The median Black household searches for 25 days longer than the median white household (Table 1. Within the voucher program, older households tend to have shorter search times, as they are often less likely to move from their initial unit. But even when we take into account differences by age, Black households still search for longer; the median Black voucher recipient over the age of 65 searches considerably longer than the median White household under 65 (Figure A4).

In Table 2, I show raw means for the dependent variables on housing turnover and neighborhood outcomes. Longer searches have lower rates of exiting and moving within 1 and 2 years. Moving to a lower poverty neighborhood shows a similar trend, though not monotonic and slightly less pronounced. Interestingly, we also see that longer searches are associating with moving to a higher poverty ZIP Code; however, this is mostly likely explained by the fact that those who search for longer are much more likely to change ZIP Codes at all.

7 Results

7.1 Effect of issuance timing

The timing of voucher issuance may affect both the probability that a voucher recipient will be successful in using their voucher, and the amount of time that they search when they are successful. In Figure 3, we see that there is no effect of the month of issuance or the interaction team for a high search time period (defined as the last week of the month), on the chance of successfully using a voucher within 1 year of issuance.[5]

However, there are strong effects on the amount of time that a household searches. Figure 4 shoes that searches that take place at the end of the month are longer than those in the middle of the month, indicating that these participants tend to roll over into an additional month of search time. An even stronger source of variation in search times is monthly variation; searches are long in the summer months, decline in September and November, and then rise again in November and December (Figure 5).

Since there is no effect on success, there is no selection into the sample of successful searches based on issuance timing. The following analysis use the month of issuance and an indicator for a high search time period as an instrument for search time, to test the effect of longer searches on neighborhood outcomes, housing turnover, and additional outcomes such as housing costs and unit size.

[5] Results are extremely similar when using a measure of lease-up success with no time window restriction.

7.2 Effects of search time on neighborhood outcomes

Longer searches may permit households to reach lower poverty neighborhood, a core but sometimes unfulfilled goal of the voucher program. In Table 3, OLS and IV regressions are shown for three different dependent variables: an indicator for moving ZIP Codes, an indicator for moving to a lower poverty ZIP Code, and an indicator for moving to a higher poverty ZIP Code. Focusing first on moving ZIP Codes at all, in the OLS model, we see that households who search for longer are more likely to move ZIP Codes. However, in the IV model, the coefficient is greatly reduced and insignificant. Intuitively, this is due to the fact that the instrumental variables approach identified the treatment effect for "compliers", or households who search for longer then their date of issuance allows them more time. A large portion of households with shorter search times do not move from their origin ZIP Code, and likely many of these household are remaining in their same home, which drives the result that longer searches are associated with a lower chance of moving ZIP Codes. However, since these households are unlikely to take up additional search time, so we see an insignificant causal effect of longer searches.

While there is no evidence that longer searches are inducing more moves, it does appear that longer searches may cause different moves among successful voucher recipients. The IV results on the probability of moving to a lower poverty ZIP Code indicate that a 50% increase in search time leads to a 2 percentage point increase in moving to a lower poverty ZIP Code, which is an increase of about 6 percent. This is is a substantial effect, considering that often much higher touch interventions are required to help households move to a broader set of neighborhoods (Bergman et al., 2024; DeLuca et al., 2023).

In the last set of models, I look at the effect on moving to a higher poverty ZIP Code. As seen in the summary statistics, longer searches are actually associated with a higher probability of moving to a higher poverty ZIP Code; this is confirmed in the OLS model. However, this is largely due to the fact that households who search for longer are more likely to move in general. When we use the IV strategy, the coefficient flips signs, reflecting the finding that longer searches help households reach lower poverty neighborhoods.

Given large difference in search times and search experiences across demographic groups and market conditions, I conduct heterogeneity tests to determine which groups are driving these effects.

Black households and households with kids have positive and significant treatment effects for moving to a lower poverty neighborhoods (Figure 6). For older households, with a household head above 65, there is no effect of longer searches, which is consistent with the fact that older households are substantially less likely to move ZIP Codes when using their voucher. Hispanic households also show no effect of longer searches.

Market conditions appear to play a large role. Table 4 shows the effects on moving to a lower poverty ZIP Code. Here, we see that effects on moving to a lower poverty ZIP Code are much larger in higher rent markets.

7.3 Effects of search time on housing turnover

Longer searches are associated with a lower change of exiting the program or moving housing units within 1 to 2 years of entering the program. This result can be driven by households finding housing that better fits their needs and goals when they search for longer. In Table 5, I show OLS and IV regressions of search time on exiting the program within 1 or 2 years. Without an instrument, there is no effect; this differs from the raw trends shown in Table 2 due to controlling for differences across individuals and markets. However, the IV regressions show negative treatment effects. Specifically, a 50% increase in search time leads to a 2 percentage point decline in exiting the program within 1 year, and a 4.8 percentage point decline in exiting within 2 years. These are substantial declines, off base rates of 8% and 18%, respectively. The average voucher holders remains in the program for 10 years; exits within 1 year of entering may be indicative of dissatisfaction with their housing unit and an inability to find another suitable unit.

Households are also more likely to stay in their same housing unit for longer when they search for longer. A 50% increase in search time causes a 2.8 percentage point decline in moving units within the following two years, a large decline as on average, around 14% of new voucher recipients will move within 2 years of their initial lease-up. This result indicates that longer searches can benefit both voucher recipients and PHAs; households enjoy more stability and lower moving costs, and housing authorities may see reduced costs as moves involve conducting additional inspections and staff resources.

Effects on program and housing unit turnover are predominately driven by Black households and households with kids, an important result given racial disparities in search times and barriers faced by households with children (Figure 7). And while effects are slightly larger in tighter rental markets, effect sizes are not significantly different (Table 7).

7.4 Additional outcomes

In addition to neighborhood mobility and match quality, there are other margins that longer searches may affect. One of these is housing costs. While low-income renters with a traditional budget constraint may try to optimize on price when given more time to search, voucher holders do not experience any added price increases up to the payment standard. In fact, given the results on moving to lower poverty neighborhoods, voucher holders may actually rent higher rent units when they search for longer. Table A1 confirms this hypothesis, showing that longer searches lead to higher gross rents, which translates into higher housing authority portions of the rent. However, there is no increase in the share of renters who have a rent burden over 30%, which indicates that voucher recipients that search longer rent more expensive units, but on average do not exceed payment standards, above which they would be obligated to pay the difference between the gross rent and the payments standard out-of-pocket.

Another margin that voucher recipients may optimize on is the size of their rental unit. When a household is issued a voucher, they qualify for a certain bedroom size based on the number, age, and sex of household members.[6] At the time of lease-up, households are generally not permitted to be "under-housed", or renting a unit with fewer bedrooms than they are qualified for, unless they receive special approval. However, households may be "over-housed", or rent a unit with more bedrooms than they are qualified for, given the unit falls within the payment standard for the smaller bedroom size that they qualified for.

Households who search for longer may attempt to increase the size of their housing unit; in this case, we would expect longer searches to cause a drop in being under-housed, and an increases in being over-housed. In Table A2, we see that there is no significant effect on either outcome, through the signs of the coefficients are in the expected direction. Table A3 shows the same outcomes, stratified by market conditions. Here, there is marginally significant increase in being over-housed, for those in low rent markets. This suggests that longer searches may allow voucher recipients to

[6]PHAs have some flexibility in setting bedroom size requirements, as long as they are not less generous than federal standards.

rent larger units, but only where rents are low enough that larger units are still affordable through the voucher program.

8 Robustness

Several robustness checks confirm the results from the prior section. First, I test a variety of different instruments. I also conduct Anderson-Rubin tests to confirm that results are robust to weak instruments. Results are also qualitatively similar when dropping 2020 and 2021, years which were affected by the COVID-19 pandemic and following rental market tightening.

9 Conclusion

The voucher program is one of our most important tools for ensuring that low-income households can find safe and affordable housing. However, the strenuous private market housing search process can prohibit families from using their vouchers, and lead to inefficient housing matches when they do use their vouchers. These results provide evidence that lengthening search times may be an effective tool in promoting better housing matches among voucher recipients.

The second component of this project involves looking at another source of variation in search times: allowable search time. Original data collection is in progress that will allow for testing the effect of different search term policies, both on success rates and on these same margins of match quality. This will provide further evidence on what PHAs stand to gain from modifying their program operations.

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Tables and figures

Figure 1: Success rates and search times, 2015-2021

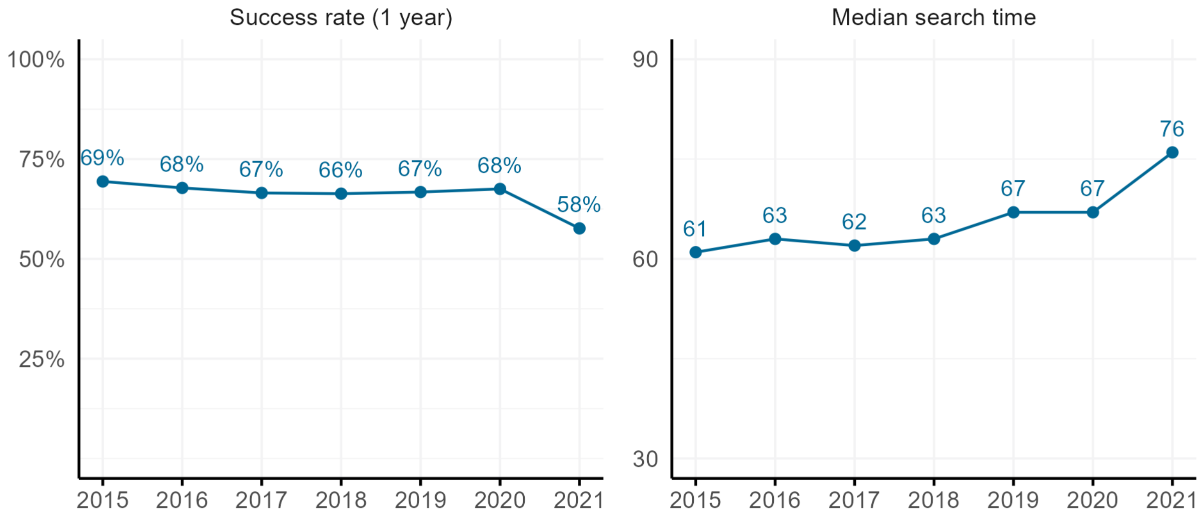


Figure 2: Distribution of success rates and search times, 2019-2020

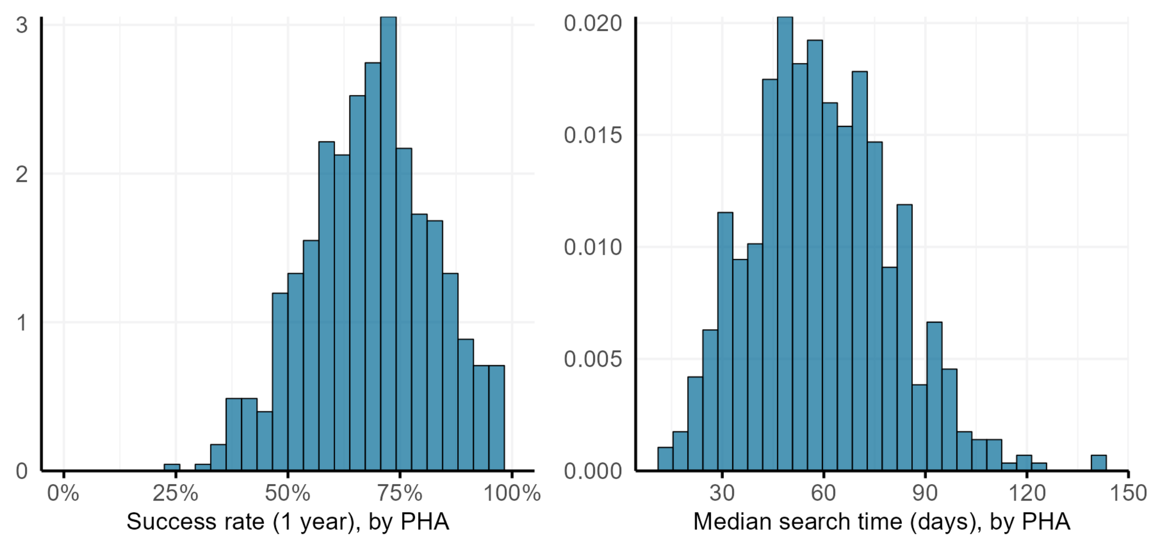


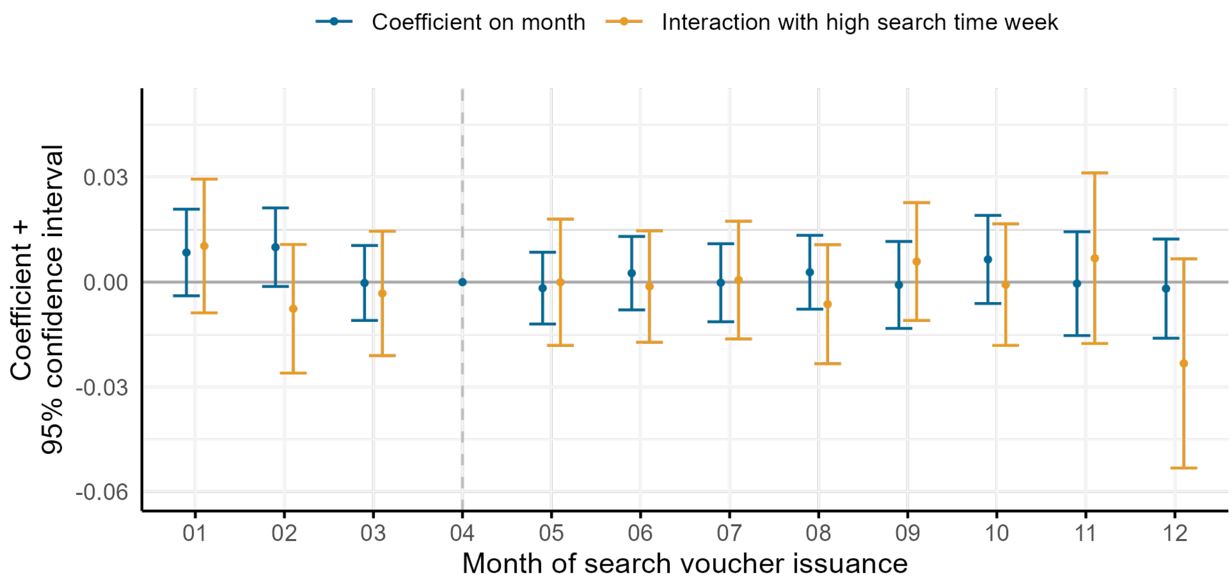
Table 1: Sample summary statistics

	Observations	Share	Success rate (1 year)	Median search time
Black head of household	393,715	0.504	0.667	75
Hispanic head of household	112,941	0.145	0.662	63
White head of household	240,571	0.308	0.648	50
Other race head of household	33,635	0.043	0.626	59
Female head of household	565,522	0.724	0.664	66
Presence of kids	389,749	0.499	0.671	69
Age 65+	79,422	0.102	0.684	47
Disability indicator	240,870	0.308	0.661	61
All observations	780,862		0.659	65

Table 2: Raw, uncontrolled outcome variables, by search time

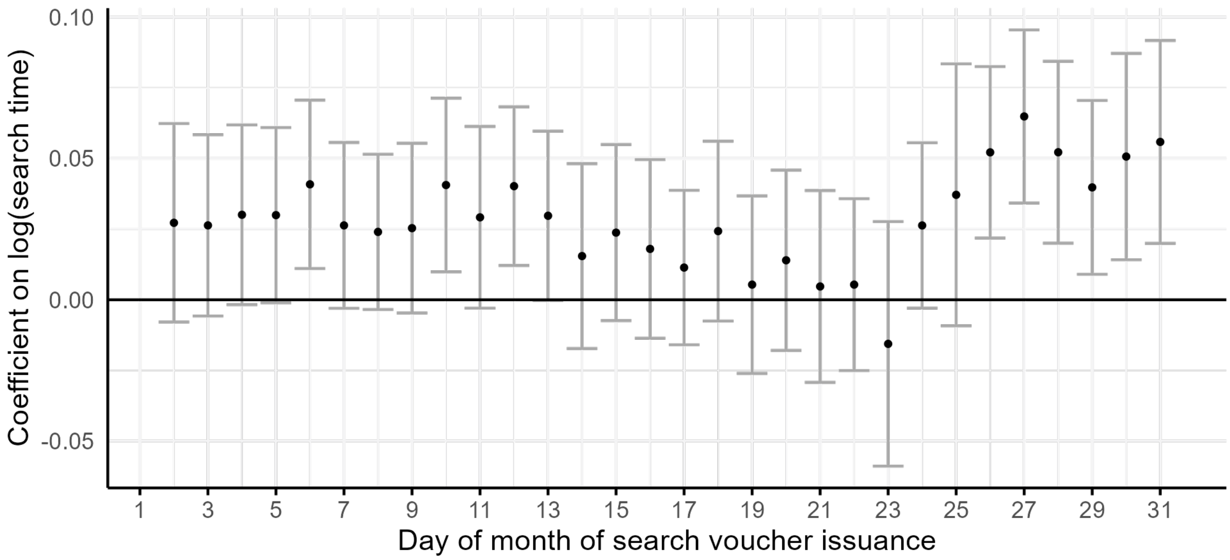
	Housing turnover				Neighborhood outcomes		
Search time	Exit within 1 year	Exit within 2 years	Move within 1 year	Move within 2 years	Move ZIP Codes	Move to lower poverty ZIP Code	Move to higher poverty ZIP Code
29 days or less	0.11	0.23	0.05	0.13	0.48	0.25	0.22
30-59 days	0.09	0.20	0.05	0.14	0.61	0.33	0.27
60-89 days	0.08	0.18	0.04	0.14	0.67	0.37	0.30
90-119 days	0.07	0.17	0.04	0.14	0.70	0.38	0.32
120+ days	0.06	0.15	0.04	0.13	0.73	0.39	0.34
All successful searches	0.08	0.19	0.05	0.14	0.64	0.34	0.29

Figure 3: Effect of issuance timing on leasing success



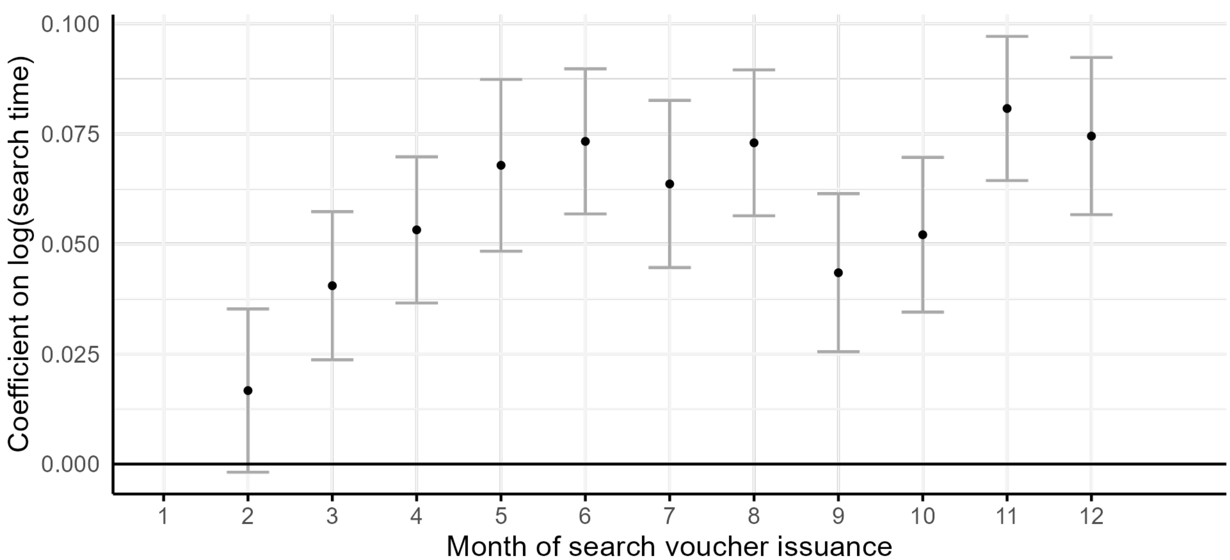
Note: Standard errors are clustered by PHA.

Figure 4: Effect of day of month on search time



Note: Standard errors are clustered by PHA.

Figure 5: Effect of month of year on search time



Note: Month of issuance and an indicator for the last week of the month are used as instruments. Standard errors are clustered by PHA.

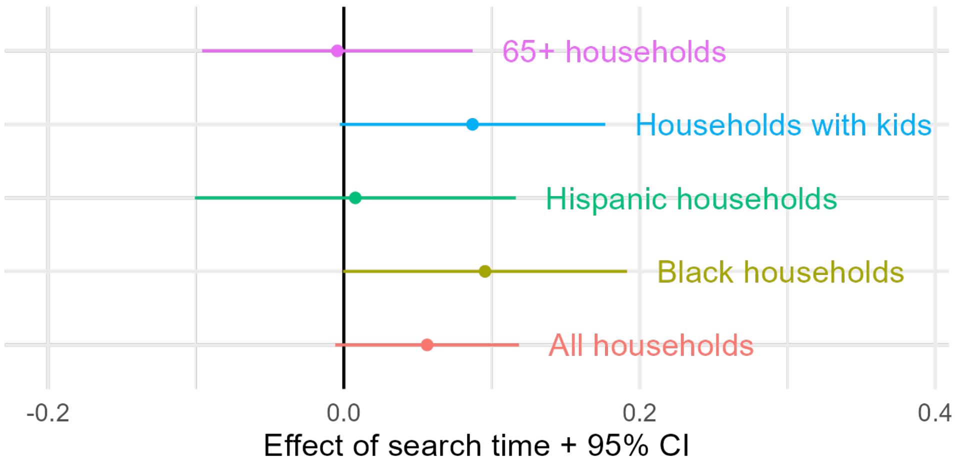
Table 3: OLS and IV estimates on moving to a lower poverty ZIP Code

	Move ZIP Codes		Move to a lower poverty ZIP Code		Move to a higher poverty ZIP Code	
	OLS	IV	OLS	IV	OLS	IV
log(Search time)	0.068***		0.032***		0.037***	
	(0.002)		(0.002)		(0.002)	
log(Search time), fitted		0.023		0.053+		−0.033
		(0.030)		(0.031)		(0.028)
Num.Obs.	432 631	432 631	424 006	424 006	424 006	424 006
PHA by Year fixed effects	X	X	X	X	X	X
Individual controls	X	X	X	X	X	X
First stage F stat		59.8		59.2		59.2

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: Month of issuance and an indicator for the last week of the month are used as instruments. Standard errors are clustered by PHA.

Figure 6: Effect of search time on moving to a lower poverty ZIP Code, stratified by demographic group



Note: Month of issuance and an indicator for the last week of the month are used as instruments. Standard errors are clustered by PHA.

Table 4: Effect of search time on moving to a lower or higher poverty ZIP Code, stratified by market rent conditions

	Move to a lower poverty ZIP Code		Move to a higher poverty ZIP Code	
	Low rent markets	High rent markets	Low rent markets	High rent markets
log(Search time), fitted	−0.015	0.145**	0.035	−0.118*
	(0.037)	(0.051)	(0.032)	(0.046)
Num.Obs.	232 556	191 450	232 556	191 450
PHA x Year fixed effects	X	X	X	X
Individual controls	X	X	X	X
First stage F stat	37.9	24	37.9	24

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Note: Month of issuance and an indicator for the last week of the month are used as instruments. Standard errors are clustered by PHA.

Table 5: OLS and IV estimates on program exits

	Exit within 1 year		Exit within 2 years	
	OLS	IV	OLS	IV
log(Search time)	0.000		0.000	
	(0.001)		(0.001)	
log(Search time), fitted		−0.039*		−0.095***
		(0.018)		(0.027)
Num.Obs.	432 631	432 631	432 631	432 631
PHA by Year fixed effects	X	X	X	X
Individual controls	X	X	X	X
First stage F stat		59.8		59.8

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

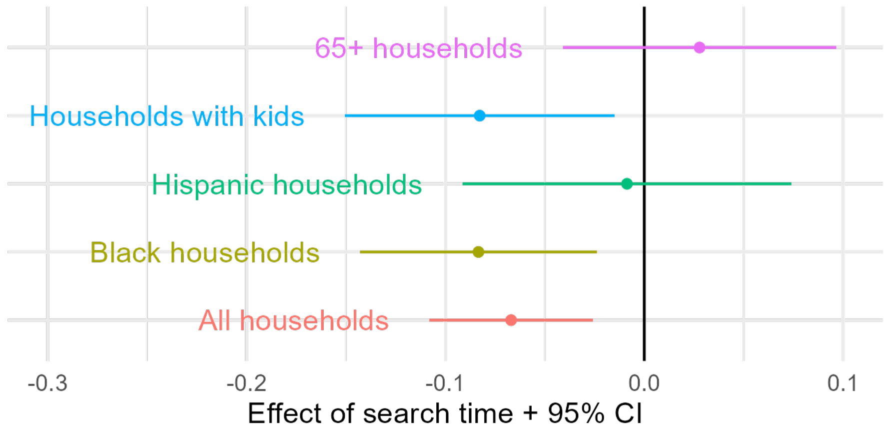
Table 6: OLS and IV estimates on moving housing units

	Move within 1 year		Move within 2 years	
	OLS	IV	OLS	IV
log(Search time)	−0.003***		−0.007***	
	(0.000)		(0.001)	
log(Search time), fitted		−0.012		−0.066**
		(0.011)		(0.021)
Num.Obs.	432 631	432 631	432 631	432 631
PHA by Year fixed effects	X	X	X	X
Individual controls	X	X	X	X
First stage F stat		59.8		59.8

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Note: Month of issuance and an indicator for the last week of the month are used as instruments. Standard errors are clustered by PHA.

Figure 7: Effect of search time on moving housing units within 2 years, stratified by individual demographics



Note: Month of issuance and an indicator for the last week of the month are used as instruments. Standard errors are clustered by PHA.

Table 7: Effect of search time on housing turnover, stratified by market rent conditions

	Exit within 2 years		Move within 2 years	
	Low rent markets	High rent markets	Low rent markets	High rent markets
log(Search time), fitted	−0.059+	−0.138**	−0.056*	−0.077*
	(0.032)	(0.046)	(0.025)	(0.034)
Num.Obs.	236 611	196 020	236 611	196 020
PHA x Year fixed effects	X	X	X	X
Individual controls	X	X	X	X
First stage F stat	38.1	24.3	38.1	24.3

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Note: Month of issuance and an indicator for the last week of the month are used as instruments. Standard errors are clustered by PHA.

Appendix

Figure A1: Issuance counts by day of month and month of year

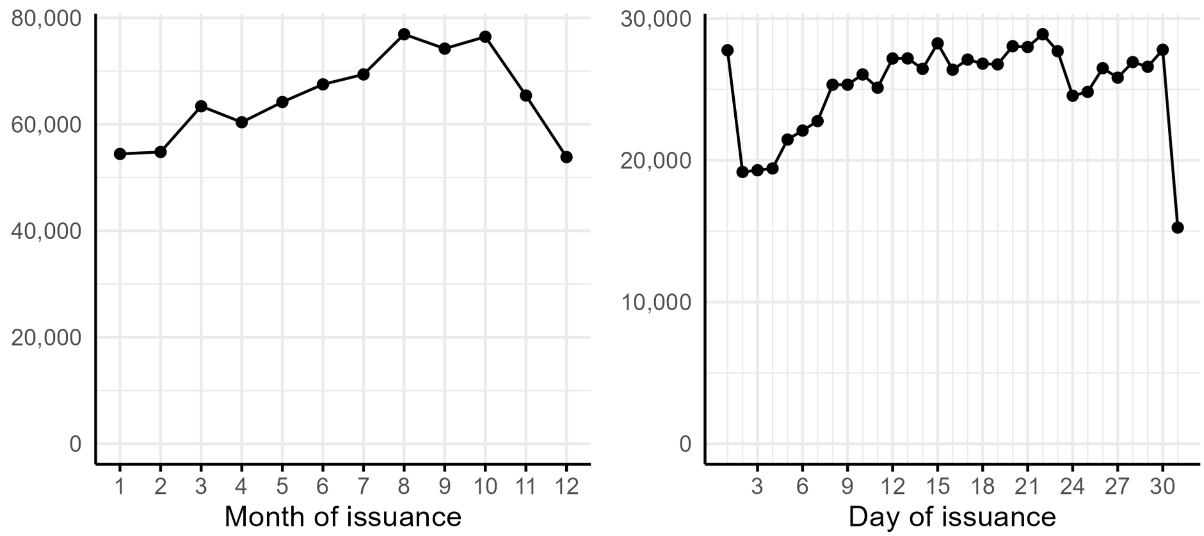


Figure A2: Balance test by day of issuance

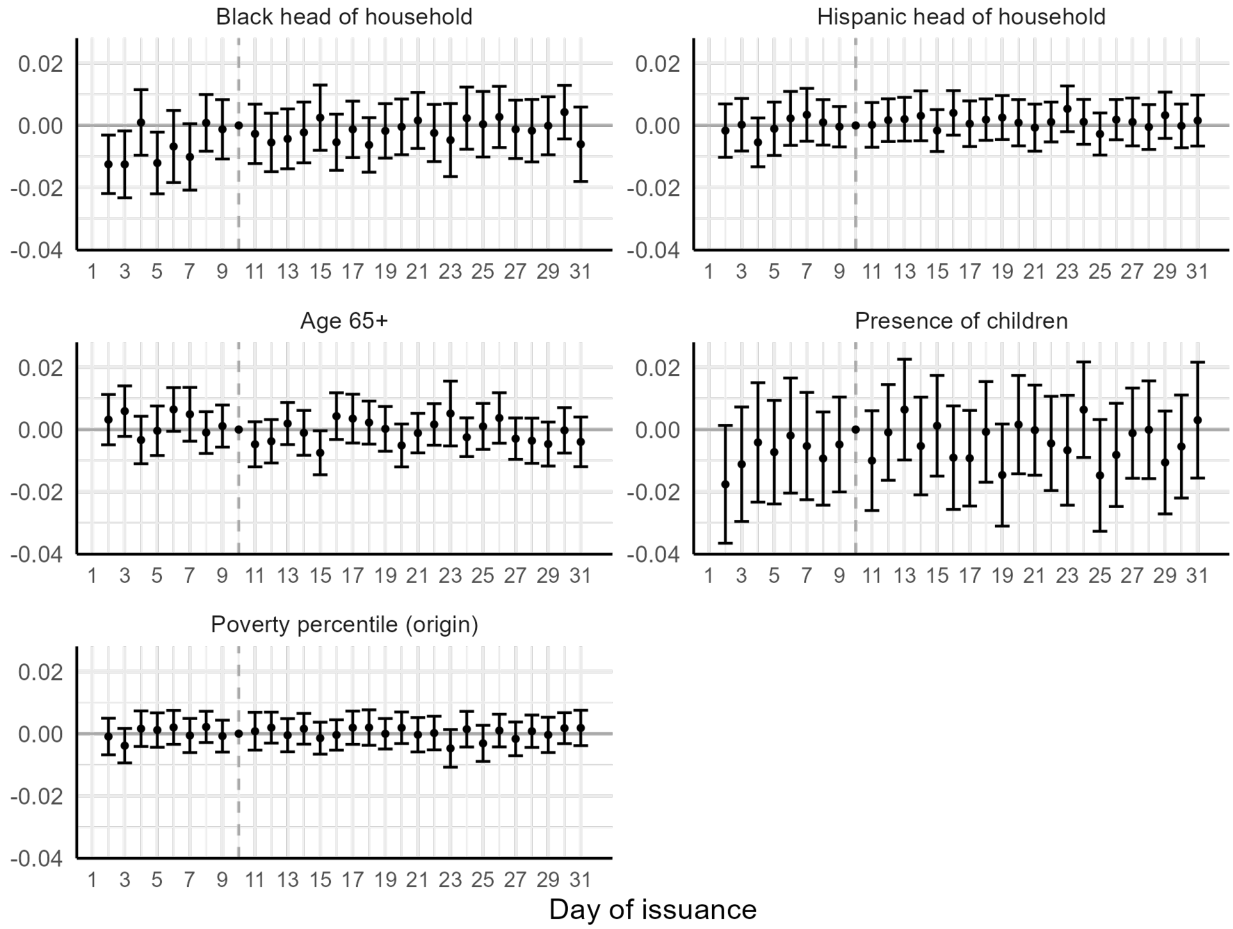


Figure A3: Balance test by month of issuance

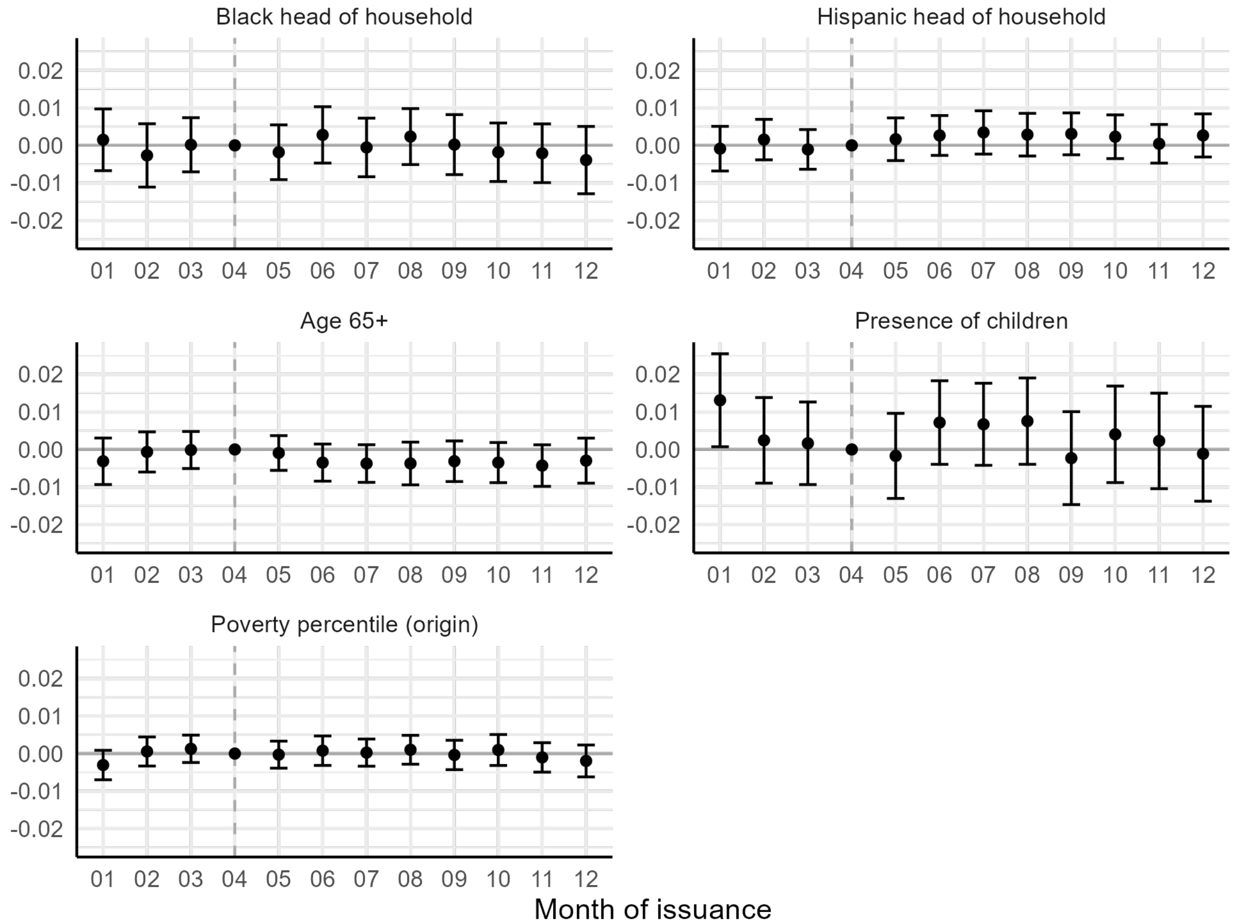


Figure A4: Median search time by race/ethnicity and age of household head

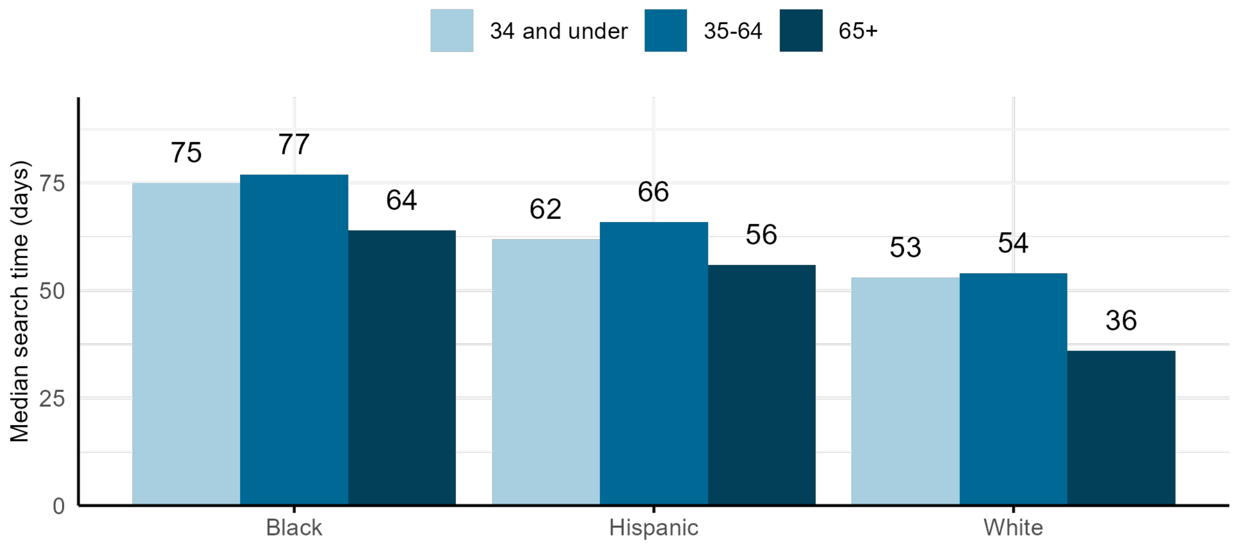


Figure A5: Median search time by county median rent quartile

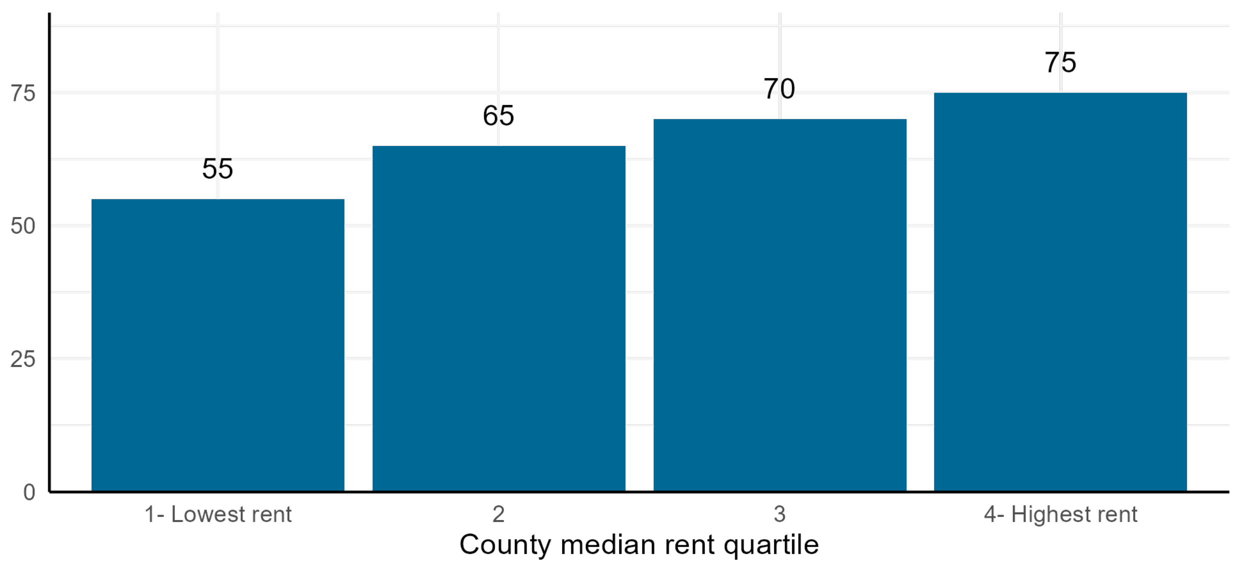


Table A1: Effect of search time on housing cost variables

	Gross rent		HAP		Rent burden over 30%	
	OLS	IV	OLS	IV	OLS	IV
log(Search time)	12.958***		22.278***		0.018***	
	(1.572)		(1.514)		(0.002)	
log(Search time), fitted		328.373***		229.429***		−0.004
		(53.259)		(42.521)		(0.034)
Num.Obs.	432 630	432 630	429 707	429 707	415 431	415 431
PHA by Year fixed effects	X	X	X	X	X	X
Individual controls	X	X	X	X	X	X
First stage F stat		59.8		59.6		59.3

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table A2: Effect of search time on relative unit size

	Under-housed		Over-housed	
	OLS	IV	OLS	IV
log(Search time)	0.010***		0.000	
	(0.002)		(0.002)	
log(Search time), fitted		−0.012		0.032
		(0.031)		(0.021)
Num.Obs.	432 615	432 615	432 615	432 615
PHA by Year fixed effects	X	X	X	X
Individual controls	X	X	X	X
First stage F stat		59.9		59.9

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table A3: Effect of search time on relative unit size, stratified by origin ZIP Code rent

	Under-housed		Over-housed	
	Low rent markets	High rent markets	Low rent markets	High rent markets
log(Search time), fitted	−0.004	−0.021	0.030+	0.042
	(0.035)	(0.051)	(0.016)	(0.048)
Num.Obs.	236 600	196 015	236 600	196 015
PHA x Year fixed effects	X	X	X	X
Individual controls	X	X	X	X
First stage F stat	38.1	24.3	38.1	24.3

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$